

COURSE CODE CSAD311

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COURSE NAME Data structure

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CO5 AT1 PSEUDO CODE

① Graph Traversal Problem - BFS and DFS

Constraints:

Avoid revisiting nodes

Track traversal order

BFS - PSEUDO CODE :

BFS (Graph, start):

Create an empty queue Q

Create a visited array and set all nodes to FALSE

Create an empty list traversal.

visited [start] = TRUE

Enqueue start into Q

while Q is not empty :

u = Dequeue Q

Add u to traversal

for each vertex v adjacent to u :

if visited [v] == FALSE :

visited [v] = TRUE

Enqueue v into Q .

return traversal

DFS Pseudo code:

DFS (Graph, start):

Create a Visited array and set all nodes to FALSE

Create an empty list traversal.

DFS - visit (start)

return traversal

DFS - visit (u):

Visited [u] = TRUE

Add u to traversal

for each vertex v adjacent to u:

if Visited [v] == FALSE

DFS - visit (v)

Time Complexity : $O(V + E)$

Space Complexity :- $O(V)$

② Social Network Analysis - BFS with Early Stopping:-

Constraints:-

Find shortest - level connection

Stop when target is found

Pseudo code:-

BFS - Early - stop (Graph, start, Target):

Create an empty Queue q

Create Visited array and set all nodes to FALSE

Create level array and set all level to -1

visited [root] = TRUE

level [root] = 0

Enqueue root into q

while q is not empty:

u = Dequeue q

if u == Target:

return level [u]

for each vertex v ^{adj} to u:

if visited [v] == FALSE:

visited [v] = TRUE

level [v] = level [u] + 1

If v == Target:

return level [v]

Enqueue v into q

return -1

Time complexity: $O(V + E)$ in worst case

Space complexity: $O(V)$

Explanation:-

* Queue ensures BFS visits nodes level by level

- * visited [] prevents revisiting the same person/nodes.

- * level [] stores the shortest number of connections from the source

- * As soon as the target is found, BFS stops

- * Therefore, the returned level represents the shortest connection path in an unweighted social network.